

Inquiry Proposal: Formal Coq Target for an Icosian-Inspired Omi-Ring

Working Title

OMI-Ring Icosian Bridge: A Coq Proof Program for Deterministic Relation Arithmetic

0. Purpose

This proposal investigates whether the **icosian ring** can serve as a formal arithmetic model for the local OMI-ring.

The immediate goal is not to prove that OMI is the Leech lattice, the Mathieu group, the Conway group, or the Monster group.

The immediate goal is smaller and stronger:

```
Can we define an OMI-ring arithmetic kernel in Coq
using golden-field quaternions,
prove closure and norm preservation,
and then connect that kernel to deterministic OMI relation movement?
```

The icosian ring is attractive because it sits between several structures already relevant to OMI:

```
golden ratio
quaternions
600-cell
icosahedral symmetry
E8 lattice
Leech lattice
Golay / Mathieu 24-coordinate structure
```

The icosian ring is known as a quaternionic construction related to the **E8** lattice; the Leech lattice can also be constructed from three copies of the icosian/E8 structure through Turyn-style construction.

For OMI, this suggests a proof ladder:

```
icosian arithmetic
→ E8-compatible local balance
```

- 24-coordinate Leech/Golay/Mathieu mental model
- OMI canon sphere-packing discipline

But the proof target must begin locally.

1. Why Icosians Matter for OMI

OMI needs a local arithmetic basis that can support:

relation movement
gauge-controlled rotation
deterministic carry-forward
bounded closure
non-hash identity
projection without authority leakage

The icosian ring may help because it is built from quaternions over the golden field.

A quaternion has the form:

$$q = a + bi + cj + dk$$

An icosian coordinate uses golden-field components:

$$a + b\sqrt{5}$$

So an icosian-like quaternion has the form:

$$\begin{aligned} &(a_0 + a_1\sqrt{5}) \\ &+ (b_0 + b_1\sqrt{5})i \\ &+ (c_0 + c_1\sqrt{5})j \\ &+ (d_0 + d_1\sqrt{5})k \end{aligned}$$

This has:

$$4 \text{ quaternion coordinates} \times 2 \text{ golden-field components} = 8 \text{ rational coordinates}$$

That is the reason the icosian ring can model an **E8**-like 8-dimensional lattice structure. The **E8** lattice is uniquely positive-definite, even, unimodular in rank 8, and it is closely related to octonionic and exceptional symmetry constructions.

OMI already uses octonionic/Fano intuition for gauge control. However, octonions are nonassociative. Quaternions are noncommutative but associative.

Therefore, the icosian route may give a more proof-friendly middle layer:

```
octonion intuition
→ quaternionic icosian proof kernel
→ E8-like local lattice
→ Leech/Golay/Mathieu 24-coordinate discipline
```

2. Core Hypothesis

The hypothesis is:

```
The Omi-Ring can be modeled, at least locally, by an icosian-inspired
quaternionic golden-field arithmetic kernel.
```

This does not mean:

```
OMI = the icosian ring
OMI = E8
OMI = the Leech lattice
OMI = M24
```

It means:

```
The icosian ring may provide a formal arithmetic witness
for the local relation basis of the Omi-Ring.
```

The useful property is not mystical symmetry.

The useful property is proof discipline:

```
closure
norm preservation
finite unit enumeration
deterministic multiplication
bounded relation stepping
```

3. Coq Scope

The Coq target should be deliberately staged.

Stage 1 — Golden Field

Define a constructive representation of the golden field:

```
Record Golden := {  
  g_re : Q;  
  g_rt : Q  
}.
```

Interpreted as:

$$g_re + g_rt\sqrt{5}$$

Define:

```
g_add : Golden -> Golden -> Golden  
g_neg : Golden -> Golden  
g_mul : Golden -> Golden -> Golden  
g_conj : Golden -> Golden  
g_norm : Golden -> Q
```

Core theorems:

```
g_add_assoc  
g_add_comm  
g_mul_assoc  
g_mul_comm  
g_mul_1_l  
g_mul_1_r  
g_distrib_l  
g_distrib_r  
g_conj_involutive  
g_norm_mul
```

Target:

Golden values form a commutative ring / field-like rational extension suitable for quaternion coordinates.

Stage 2 — Quaternions over the Golden Field

Define quaternions:

```
Record GQuat := {  
  q_r : Golden;  
  q_i : Golden;  
  q_j : Golden;  
  q_k : Golden  
}.
```

Define Hamilton multiplication:

```
i2 = j2 = k2 = ijk = -1  
ij = k, ji = -k  
jk = i, kj = -i  
ki = j, ik = -j
```

Core operations:

```
q_add : GQuat -> GQuat -> GQuat  
q_neg : GQuat -> GQuat  
q_mul : GQuat -> GQuat -> GQuat  
q_conj : GQuat -> GQuat  
q_norm : GQuat -> Golden
```

Core theorems:

```
q_add_assoc  
q_add_comm  
q_mul_assoc  
q_mul_1_l  
q_mul_1_r  
q_distrib_l  
q_distrib_r  
q_conj_involutive  
q_norm_mul
```

Important: quaternion multiplication is not commutative, so do not try to prove:

```
q_mul_comm
```

The useful fact is associativity.

That is why this is easier than octonions.

Stage 3 — Unit Icosian Candidates

Define the 120 unit icosians as an explicit finite list.

Conceptually, these are the quaternionic vertices of the 600-cell:

```
1/2(±2, 0, 0, 0)
1/2(±1, ±1, ±1, ±1)
1/2(0, ±1, ±1/φ, ±φ)
```

where permutations and sign rules generate 120 units.

The 120 unit icosians form the binary icosahedral group, realized as vertices of the 600-cell.

Coq representation:

```
Definition unit_icosians : list GQuat := ...
```

Predicates:

```
Definition IsUnitIcosian (x : GQuat) : Prop :=
  In x unit_icosians.
```

Core theorems:

```
unit_icosians_length :
  length unit_icosians = 120.

unit_icosians_norm_one :
  forall x, IsUnitIcosian x -> q_norm x = 1.

unit_icosians_closed_mul :
  forall x y,
```

```

    IsUnitIcosian x ->
    IsUnitIcosian y ->
    IsUnitIcosian (q_mul x y).

unit_icosians_has_identity :
  IsUnitIcosian q_one.

unit_icosians_has_inverse :
  forall x,
    IsUnitIcosian x ->
    IsUnitIcosian (q_conj x).

```

This proves that the unit icosians form a finite multiplicative group.

Stage 4 — Icosian Ring Candidate

Define the icosian ring as the additive span generated by unit icosians.

For a first Coq milestone, do not attempt a full infinite ring quotient.

Use a finite-support representation:

```

Record IcosianSpan := {
  coeffs : list (Z * GQuat);
  coeffs_are_units :
    forall z q, In (z,q) coeffs -> IsUnitIcosian q
}.

```

Then define normalization later.

Initial target theorems:

```

span_closed_add :
  forall x y,
    IsIcosianSpan x ->
    IsIcosianSpan y ->
    IsIcosianSpan (span_add x y).

span_closed_neg :
  forall x,
    IsIcosianSpan x ->
    IsIcosianSpan (span_neg x).

span_closed_mul :
  forall x y,

```

```
IsIcosianSpan x ->
IsIcosianSpan y ->
IsIcosianSpan (span_mul x y).
```

This gives the first formal notion of “ring closure.”

Stage 5 — Norm / Balance Law

The OMI reason to care about the icosian ring is not just closure.

It is balance.

Define a norm-like measure that is compatible with OMI relation movement.

For quaternions:

```
N(q) = q * conjugate(q)
```

For OMI:

```
norm preservation = relation movement does not destroy local balance
```

Target theorem:

```
omi_icosian_norm_mul :
  forall x y,
    q_norm (q_mul x y) = g_mul (q_norm x) (q_norm y).
```

Then specialize:

```
unit_step_preserves_norm :
  forall state step,
    IsUnitIcosian step ->
      q_norm (q_mul state step) = q_norm state.
```

This is the key bridge.

If OMI gauge steps are interpreted as unit icosian actions, then applying a gauge step preserves norm.

That gives a possible formal basis for:

gauge-controlled relation movement preserves local balance

Stage 6 — OMI-Ring Interface

Only after the arithmetic kernel is proven should we define the OMI interface.

Define:

```
Record OmiRingState := {  
  ors_value : GQuat;  
  ors_valid : IsIcosianLike ors_value  
}.
```

Define gauge action:

```
Definition OmiGauge := Fin.t 16.  
Definition gauge_to_unit : OmiGauge -> GQuat.
```

Then:

```
Definition omi_ring_step  
  (s : OmiRingState)  
  (g : OmiGauge)  
  : OmiRingState :=  
  multiply_by_unit s (gauge_to_unit g).
```

Core theorems:

```
omi_ring_step_closed :  
  forall s g,  
    IsOmiRingState s ->  
    IsOmiRingState (omi_ring_step s g).  
  
omi_ring_step_deterministic :  
  forall s g x y,  
    omi_ring_step s g = x ->  
    omi_ring_step s g = y ->  
    x = y.  
  
omi_ring_step_preserves_norm :
```

```
forall s g,  
  omi_norm (omi_ring_step s g) = omi_norm s.
```

This is the actual OMI theorem target.

Stage 7 — Relation Quotation Compatibility

Now connect to OMI doctrine.

OMI-Lisp does not execute.

It declares candidate relations.

Relational quotation frames the candidate relation.

The Omi-Ring arithmetic may be used to model candidate motion, but does not itself accept state.

Define:

```
Record QuotedRelation := {  
  qr_source : OmiRingState;  
  qr_gauge   : OmiGauge;  
  qr_target  : OmiRingState;  
  qr_claim   : qr_target = omi_ring_step qr_source qr_gauge  
}.
```

The Tetragrammatron adjudicates whether the claim is valid.

Target theorem:

```
quoted_relation_adjudicable :  
  forall qr,  
    Decidable (qr_target qr = omi_ring_step (qr_source qr) (qr_gauge qr)).
```

This is the formal version of:

```
candidate declaration can be deterministically adjudicated
```

4. What This Does Not Prove Yet

This proposal does not yet prove:

```
OMI is the Leech lattice.  
OMI has Conway group symmetry.  
OMI realizes M24.  
OMI constructs the Monster group.
```

Those are later and much larger targets.

The immediate proof target is smaller:

```
a local OMI-ring arithmetic kernel has closure,  
norm preservation,  
and deterministic gauge action.
```

Once that is proven, we can ask whether the 24-file canon spine, Golay/Mathieu structure, or Leech lattice mental model can be formalized on top.

5. Expected Repo Placement

This should live in `omi-axioms`.

Suggested Coq module path:

```
coq/GoldenField.v  
coq/GoldenQuaternion.v  
coq/IcosianUnits.v  
coq/IcosianRing.v  
coq/OmiRingIcosian.v  
coq/OmiRingQuotation.v
```

Suggested dependency order:

```
GoldenField  
  ↓  
GoldenQuaternion  
  ↓  
IcosianUnits  
  ↓  
IcosianRing  
  ↓  
OmiRingIcosian
```

↓
OmiRingQuotation

Suggested export bridge:

```
Require Export OmiRingIcosian.  
Require Export OmiRingQuotation.
```

6. Haskell Support

Although the target is Coq first, Haskell can still support the proof program.

Haskell can be used to:

```
enumerate the 120 unit icosians  
generate multiplication tables  
test closure  
test norm preservation  
export tables as Coq definitions
```

The Haskell side should not be treated as proof authority.

It is a generator, explorer, and regression oracle.

The proof authority remains Coq.

Suggested Haskell modules:

```
OMI.Icosian.Golden  
OMI.Icosian.Quaternion  
OMI.Icosian.Units  
OMI.Icosian.Tables  
OMI.Icosian.ExportCoq
```

7. Why This Is a Good First Formal Target

This target is good because it is:

```
local  
bounded  
arithmetic
```

deterministic
compatible with OMI gauge control
more proof-friendly than octonions
connected to E8 / Leech / Mathieu structure later

It does not require proving a huge sporadic-group theorem.

It gives OMI a formal arithmetic kernel first.

That matches the project discipline:

prove the local relation law
then build outward

8. Canonical Inquiry Statement

The icosian ring is investigated as a candidate arithmetic model for the Omi-Ring local relation basis.

The first formal target is a Coq proof that golden-field quaternionic units are closed under multiplication, preserve norm, and induce deterministic OMI gauge steps.

This proof does not make the icosian ring a protocol authority.

It provides a possible arithmetic witness for local relation movement.

OMI identity remains the addressed place-value relation.

Validation determines.

Omi-Attestation witnesses.

Accepted Omi-State may be recorded.

Projection only displays accepted relation state.

9. First Milestone

The first milestone should be:

GoldenField.v + GoldenQuaternion.v compile,
and Coq proves quaternion multiplication is associative
and norm is multiplicative over Golden.

Only after that should we enumerate the 120 unit icosians.

10. Second Milestone

The second milestone should be:

```
IcosianUnits.v defines the 120 units  
and proves closure under multiplication  
using either direct finite-table verification  
or symbolic quaternion/golden arithmetic.
```

11. Third Milestone

The third milestone should be:

```
OmiRingIcosian.v defines 16 gauge actions as selected unit actions  
and proves deterministic, closed, norm-preserving OMI-ring stepping.
```

That is the first proof that directly matters for OMI.